



BAULKHAM HILLS HIGH SCHOOL

2016

YEAR 12 June - Task 3

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work
- Attempt all questions
- Start a new page for each question

Total marks – 34

Exam consists of 4 pages.

Standard integrals provided on page 5

Question 1 (11 marks) Start on the appropriate page of your answer booklet.**Marks**

a) Find the indefinite integrals:

$$\text{i) } \int \frac{dx}{\sqrt{x^2 + 6x + 13}} \quad 2$$

$$\text{ii) } \int \frac{x+3}{x^2+9} dx \quad 2$$

$$\text{iii) } \int \frac{\sin^3 x}{\cos^2 x} dx \quad 3$$

b) i) Find the numbers a, b and c such that

$$\frac{x-6}{(x+4)(x-1)^2} = \frac{a}{x+4} + \frac{b}{x-1} + \frac{c}{(x-1)^2}$$

ii) Hence find

$$\int \frac{x-6}{(x+4)(x-1)^2} dx$$

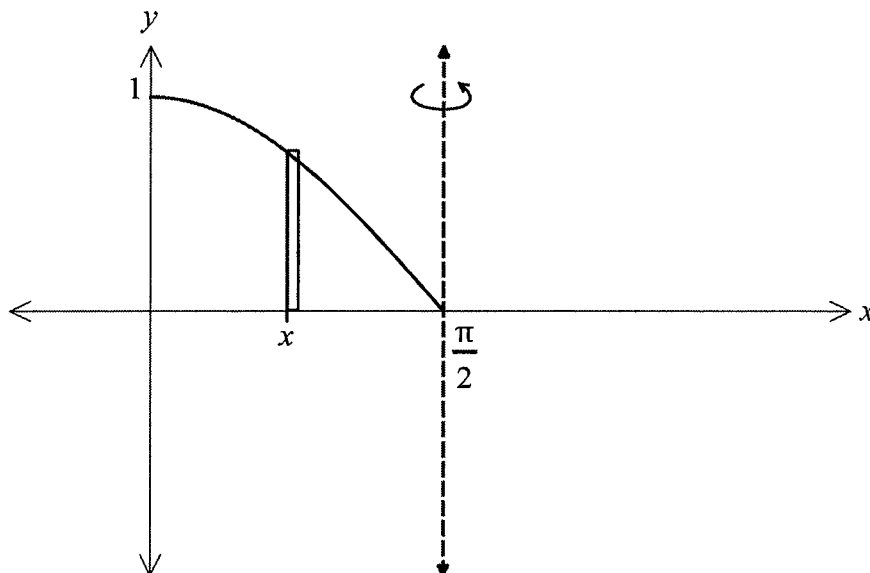
2**2**

Question 2 (13 marks) Start on the appropriate page of your answer booklet.**Marks**

a) Evaluate:

4

$$\int_1^{\sqrt{3}} \frac{dx}{x\sqrt{1+x^2}}$$

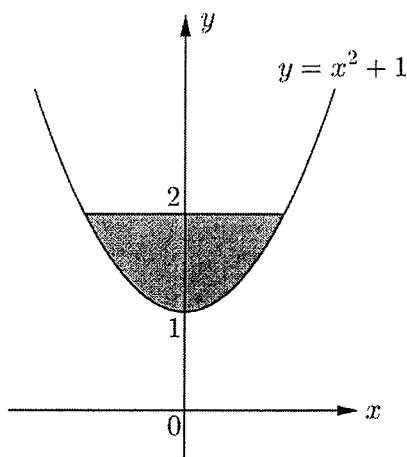
b) i) Prove $\int_0^a f(x)dx = \int_0^a f(a-x)dx$ **2**ii) Hence evaluate $\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$.**3**c) The region in the first quadrant bounded by $y = \cos x$, x -axis and the y -axis is rotated about the line $x = \frac{\pi}{2}$.i) Using the method of cylindrical shells, show that the volume, ΔV , of a typical shell at a distance x from the origin and thickness Δx is given by**1**

$$\Delta V = 2\pi \left(\frac{\pi}{2} - x \right) (\cos x) \Delta x$$

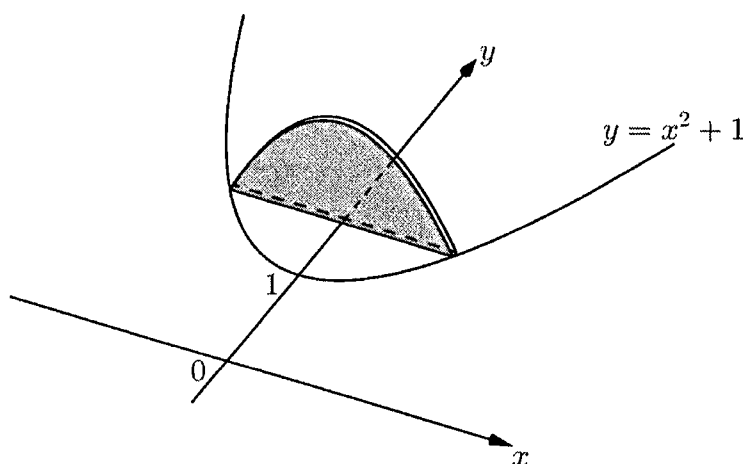
ii) Hence find the volume of this solid.

3

- a) The base of a solid, S , is the region enclosed by the parabola $y = x^2 + 1$ and the line $y = 2$.



Each cross section of S , perpendicular to the y -axis is another parabola. The height of each cross section that is y units from the origin is $\frac{1}{2}y$ units.



- (i) Using Simpson's Rule, show that the area of each cross section y units from the origin is given by $\frac{2}{3}y\sqrt{y-1}$ square units. 2
- (ii) Hence find the volume of the solid S . 3
- b) If $I_n = \int_0^1 x^n (x-1)^6 dx$ for $n \geq 0$
- (i) Show that $I_n = \frac{n}{n+7} I_{n-1}$ for $n \geq 1$ 3
- (ii) Hence find $\int_0^1 x^3 (x-1)^6 dx$ 2

X2 - 2016 June - Answers

Question 1

a) i) $\int \frac{dx}{\sqrt{x^2+6x+13}} = \int \frac{dx}{\sqrt{x^2+6x+9+4}} = \int \frac{dx}{\sqrt{(x+3)^2+2^2}} \quad \textcircled{1}$

$= \ln(x+3 + \sqrt{(x+3)^2+4}) = \ln(x+3 + \sqrt{x^2+6x+13}) + C \quad \textcircled{1}$

ii) $\int \frac{x+3}{x^2+9} dx = \int \frac{x}{x^2+9} + \frac{3}{x^2+9} dx$

$= \frac{1}{2} \ln(x^2+9) + \tan^{-1} \frac{x}{3} + C \quad \textcircled{1}$

iii) $\int \frac{\sin^3 x dx}{\cos^2 x} = \int \frac{(1-\cos^2 x) \sin x dx}{\cos^2 x} = \int \frac{\sin x}{\cos^2 x} - \sin x dx$

$= \int \tan x \cdot \sec x dx - \int \sin x dx$ 3-correct solns.

$= \sec x + \cos x + C$ 2-correct approach and one of the integrand correct

OR $\frac{\sin^2 x}{\cos x} + 2 \sec x + C$ (using IBP) 1-significant progress towards solns.

OR $\frac{x-b}{(x+4)(x-1)^2} = \frac{a}{x+4} + \frac{b}{x-1} + \frac{c}{(x-1)^2}$

$x-b = a(x-1)^2 + b(x+4)(x-1) + c(x+4)$

$x=1 \therefore -5 = a(5) \therefore a = -1$ 2-all answers correct

$a = -2/5$

$x = -4 \therefore -10 = a(25) \therefore a = -2/5$ 2-few of the three answers correct

$b = 2/5$

correct

Q1.b) ii) $\int \frac{x-b}{(x+4)(x-1)^2} dx = \int \frac{-2/5}{x+4} + \frac{2/5}{x-1} - \frac{1}{(x-1)^2} dx$

$= -\frac{2}{5} \ln(x+4) + \frac{2}{5} \ln(x-1) - \frac{(x-1)^{-1}}{-1} + C$

$= \frac{2}{5} \ln \frac{x-1}{x+4} + \frac{1}{x-1} + C$

2-correct soln
1-applying part, correctly

Question 2 (13 marks)

a) $\int_1^{\sqrt{3}} \frac{dx}{x(1+x^2)}$ let $x = \tan \theta$
 $1+x^2 = \sec^2 \theta$
 $\frac{dx}{d\theta} = \sec^2 \theta$

$= \int_{\pi/4}^{\pi/3} \frac{\sec^2 \theta d\theta}{\tan \theta \cdot \sec^2 \theta} \quad \textcircled{1} \quad x=1 \therefore \theta = \frac{\pi}{4}$

$= \int_{\pi/4}^{\pi/3} \frac{1}{\tan \theta} d\theta \quad x=\sqrt{3} \quad \theta = \frac{\pi}{3}$

$= \int_{\pi/4}^{\pi/3} \sec \theta d\theta = \int_{\pi/4}^{\pi/3} \csc \theta d\theta \quad \textcircled{1}$

$= \left[-\ln(\csc \theta + \cot \theta) \right]_{\pi/4}^{\pi/3} \quad \textcircled{1}$

$= -\ln \left[\frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}} \right] + \ln \left[\sqrt{2} + 1 \right]$

$\frac{3}{\sqrt{3}} = \sqrt{3}$

$= \ln \frac{\sqrt{2}+1}{\sqrt{3}}$

4-correct solns
3-correct subst.
and limits
simplifying to cosec and progress towards solution
2-correct subst. & limits and simpl. to cosec or progress towards solution
1-correct subst. and limits

8.2 b) i) $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

RHS = $\int_0^a f(a-x) dx$ let $u = a-x$
 $du = -dx$
 $x=0 \therefore u=a$
 $x=a \therefore u=0$

= $\int_a^0 f(u) \cdot (-du)$

= $\int_0^a f(u) du$ 2 marks - correct proof
 1 mark - correct subst and limits

= $\int_0^a f(u) du = \int_0^a f(x) dx = \text{LHS} \therefore \text{proven}$

ii) $\int_0^\pi \frac{x \sin x}{1+\cos^4 x} dx = \int_0^\pi \frac{(\pi-x) \cdot \sin(\pi-x)}{1+\cos^4(\pi-x)} dx$

= $\int_0^\pi \frac{x \sin x}{1+\cos^4 x} dx = \int_0^\pi \frac{(\pi-x) \cdot \sin x}{1+(-\cos x)^2} dx$

$\therefore I = \int_0^\pi \frac{x \sin x}{1+\cos^4 x} dx = \int_0^\pi \frac{\pi \sin x}{1+\cos^4 x} dx - \int_0^\pi \frac{x \sin x}{1+\cos^4 x} dx$ (1)

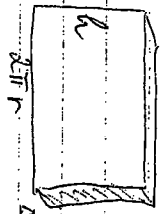
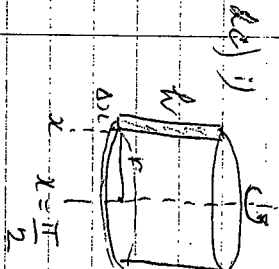
$\therefore 2I = \pi \int_0^\pi \frac{\sin x}{1+\cos^4 x} dx$ (1)

$\therefore 2I = \pi \int_{-1}^1 \frac{-du}{1+u^2} = \pi \int_{-1}^1 \frac{du}{1+u^2}$
 let $u = \cos x$
 $du = -\sin x dx$
 $x=0 \therefore u=1$
 $x=\pi \therefore u=-1$

$2I = \pi \left[\tan^{-1} u \right]_{-1}^1 = \pi \left[\frac{\pi}{4} - \left(-\frac{\pi}{4} \right) \right] = \frac{\pi^2}{2}$

2 b ii) cont:

$\int_0^\pi \frac{x \sin x}{1+\cos^2 x} dx = \frac{\pi^2}{4}$



where $h=y=\cos x$
 $r = \frac{\pi}{2} - x$

$\therefore \Delta V = 2\pi r h \Delta x$
 $\Delta V = 2\pi \left(\frac{\pi}{2} - x \right) (\cos x) \Delta x$ (1)

ii) $V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^{\pi/2} 2\pi \left(\frac{\pi}{2} - x \right) (\cos x) \Delta x$

$\therefore V = 2\pi \int_0^{\pi/2} \left(\frac{\pi}{2} - x \right) \cos x dx$

$\therefore V = 2\pi \left[\frac{\pi}{2} \sin x \right]_0^{\pi/2} - 2\pi \int_0^{\pi/2} x \cos x dx$
 3 - correct solns.
 2 - correctly integrates by parts
 1 - significant progress towards solns.

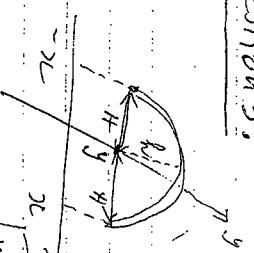
$\therefore V = \pi^2 \left[1 - 0 \right] - 2\pi \left[x \sin x \right]_0^{\pi/2} + 2\pi \int_0^{\pi/2} \sin x dx$
 $= \pi^2 - 2\pi \left[\frac{\pi}{2} - 0 \right] + 2\pi \left[-\cos x \right]_0^{\pi/2} = -2\pi \left[0 - 1 \right] = 2\pi$ (1)

3 - correct solns

2 - correctly applying the rule and finding a correct integral for 2I
 1 - correctly applying the rule and simplifying the integral

Question 3:

a)



Each cross-section = parabola

i) Area parabola = $\frac{H}{3} (0 + 4 + \frac{1}{2}y + 0)$ ①

where $h = \frac{1}{2}y$ and $H = x$ ①

Area parabola = $\frac{x}{3} \times 2y$ length of each strip

but $y = x^2 + 1$ $\therefore x = \sqrt{y-1}$ but $H > 0$

$\therefore$ Area parabola = $\frac{\sqrt{y-1}}{3} \times 2y = \frac{2}{3} y \sqrt{y-1}$ $\therefore$ Show

ii) $\Delta V = \text{Area cross-section} \times \Delta y = \frac{2}{3} y \sqrt{y-1} \cdot \Delta y$

$V = \lim_{\Delta y \rightarrow 0} \sum_{y=1}^2 \frac{2}{3} y \sqrt{y-1} \cdot \Delta y = \frac{2}{3} \int_1^2 y \sqrt{y-1} dy$

$V = \frac{2}{3} \int_1^2 y \sqrt{y-1} dy$ let $u = y-1$ $\therefore du = dy$

$V = \frac{2}{3} \int_0^1 (u+1) \sqrt{u} du$ ① OR $\frac{2}{3} \int_0^1 (y-1) \sqrt{y-1} + \sqrt{y-1} dy$

$V = \frac{2}{3} \int_0^1 u^{\frac{3}{2}} + u^{\frac{1}{2}} du = \frac{2}{3} \left[\frac{2u^{\frac{5}{2}}}{5} + \frac{2u^{\frac{3}{2}}}{3} \right]_0^1$

$= \frac{2}{3} \left[\frac{2}{5} + \frac{2}{3} \cdot 0 \right] = \frac{32}{45} \text{ (units}^3\text{)}$

OR by disc $\frac{2}{3} \int_0^1 \pi (u+1)^2 du = \frac{2}{3} \pi \int_0^1 (u^2 + 2u + 1) du$

Marks

3a)

3

correct solns:

2

deriving the volume formula and progress towards solns

1

correctly derives the volume formula

3b)

i) show $I_n = \frac{n}{n+7} I_{n-1}$ if $I_n = \int_0^1 x(x-1)^n dx$

soln: $I_n = \int_0^1 x(x-1)^n dx$ let $u = x-1$ $u' = 1$ $x = u+1$ $V = (x-1)^7$

$I_n = \int_0^1 (u+1) u^n du = \int_0^1 u^n du + \int_0^1 u^{n+1} du$

$I_n = \frac{1}{n+1} - \frac{1}{n+2}$ ①

$I_n = \frac{1}{n+1} - \frac{1}{n+2}$

$I_n = \frac{1}{n+1} - \frac{1}{n+2}$ ①

$I_n = \frac{1}{n+1} - \frac{1}{n+2}$

$\therefore I_n + \frac{1}{n+2} I_n = \frac{1}{n+2} \int_0^1 (x-1)^n dx$

$I_n \left(1 + \frac{1}{n+2} \right) = \frac{1}{n+2} \int_0^1 x(x-1)^n dx$ ①

$\therefore I_n \left(\frac{n+3}{n+2} \right) = \frac{1}{n+2} I_{n+1}$

$I_n = \frac{n}{n+2} I_{n+1}$ $\therefore I_n = \frac{n}{n+2} I_{n+1}$ $\therefore$ Show

Question 3b i)

3 - correct solns.

2 - obtains $I_n = -\frac{n}{7} \int_0^1 x^{n-1} (x-1)^7 dx$
and significant progress towards solns.

1 - Attempts to use integration by parts,
or equivalent merit.

3b ii) $I_3 = \frac{3}{7+3} I_{3-1} = \frac{3}{10} I_2 = \frac{3}{10} \times \frac{2}{9} I_1$

$\therefore I_3 = \frac{3}{10} \times \frac{2}{9} \times \frac{1}{8} I_0$ (1)

$I_3 = \frac{1}{120} \int_0^1 x^0 (x-1)^6 dx$

$\therefore I_3 = \frac{1}{120} \int_0^1 \frac{(x-1)^2}{1} dx = \frac{1}{120} \int_0^{-1} \frac{-1}{-7} dx$

$I_3 = \frac{1}{840}$ (1)

2 marks - correct soln.

1 mark - correctly applies the recurrence rule

Question 1

$$a) i) \int \frac{dx}{\sqrt{x(x-4)}} = \int \frac{dx}{\sqrt{(x-2)^2-4}} \quad 1$$

$$= \ln |(x-2) + \sqrt{(x-2)^2-4}| + C$$

$$= \ln |(x-2) + \sqrt{x^2-4}| + C \quad 1$$

$$ii) \text{ Let } u = \sin^{-1}x$$

$$\frac{du}{dx} = \frac{1}{\sqrt{1-x^2}}$$

$$\int \frac{e^{\sin^{-1}x}}{\sqrt{1-x^2}} dx = \int e^u du \quad 1$$

$$= e^{\sin^{-1}x} + C \quad 1$$

$$iii) \int \frac{1}{1+e^x} dx = \int 1 - \frac{e^x}{1+e^x} dx \quad 1$$

$$= x - \ln(1+e^x) + C \quad 1$$

$$b) \text{ Let } t = \tan \frac{x}{2}$$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$$

$$= \frac{1}{2} (1 + \tan^2 \frac{x}{2})$$

$$= \frac{1}{2} (1+t^2)$$

$$\therefore dx = \frac{2dt}{1+t^2}$$

$$x=0, t=0$$

$$x=\pi, t=1$$

$$\therefore \int_0^{\pi} \frac{dx}{2-\sin x+2\cos x} = \int_0^1 \frac{2dt}{(1+t^2)(2-\frac{2t}{1+t^2}+2\frac{(1-t^2)}{1+t^2})} \quad 1$$

$$= \int_0^1 \frac{1}{2-t} dt$$

$$= [-\ln(2-t)]_0^1 \quad 1$$

$$= -\ln 1 + \ln 2$$

$$= \ln 2 \quad 1$$

Question 2

a (i) $3x^2 - 4x + 3 \equiv A(x^2 - x + 2) + (Bx + C)(x - 1)$

sub. $x = 1$, $3 - 4 + 3 = A(1 - 1 + 2)$

$$A = 1$$

Equating coefficients of x^2

$$A + B = 3$$

$$1 + B = 3$$

$$B = 2$$

Constant term

$$2A - C = 3$$

$$2 - C = 3$$

$$C = -1$$

2 marks for correct answers

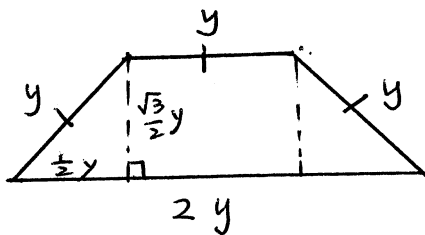
1 mark for finding two correct values.

(ii) $I = \int \frac{1}{x-1} dx + \int \frac{2x-1}{x^2-x+2} dx$

$$= \ln(x-1) + \ln(x^2-x+2) + C$$

$\frac{1}{\quad} \qquad \frac{1}{\quad}$

(b) (i)



$$A = \frac{1}{2}(y + 2y) \frac{\sqrt{3}}{2}y$$

$$= \frac{3\sqrt{3}y^2}{4}$$

$$= \frac{3\sqrt{3}}{4} \ln(x+1) \quad \frac{1}{\quad}$$

$$\Delta V = A \Delta x$$

$$= \frac{3\sqrt{3}}{4} \ln(x+1) \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^3 \frac{3\sqrt{3}}{4} \ln(x+1) \Delta x$$

$$= \frac{3\sqrt{3}}{4} \int_0^3 \ln(x+1) dx \quad \frac{1}{\quad}$$

ii) Let $u = x+1$

$$\frac{du}{dx} = 1$$

$$x=0, u=1$$

$$x=3, u=4$$

$$\begin{aligned} \therefore V &= \frac{3\sqrt{3}}{4} \int_1^4 \ln u \, du \\ &= \frac{3\sqrt{3}}{4} \left\{ [u \ln u]_1^4 - \int_1^4 1 \, du \right\} \\ &= \frac{3\sqrt{3}}{4} \{ (4 \ln 4 - 4) - (0-1) \} \\ &= \frac{3\sqrt{3}}{4} (4 \ln 4 - 3) \\ &= \frac{3\sqrt{3}}{4} (8 \ln 2 - 3) \text{ cubic units} \end{aligned}$$

Question 3

$$\begin{aligned} \text{(a)} \quad \int_0^\pi \frac{x \sin^3 x}{1 + \cos^2 x} \, dx &= \int_0^\pi \frac{(\pi-x) \sin^3(\pi-x)}{1 + \cos^2(\pi-x)} \, dx \\ &= \int_0^\pi \frac{(\pi-x) \sin^3 x}{1 + \cos^2 x} \, dx \\ &= \int_0^\pi \frac{\pi \sin^3 x}{1 + \cos^2 x} \, dx - \int_0^\pi \frac{x \sin^3 x}{1 + \cos^2 x} \, dx \end{aligned}$$

$$2 \int_0^\pi \frac{x \sin^3 x}{1 + \cos^2 x} \, dx = \int_0^\pi \frac{\pi \sin^3 x}{1 + \cos^2 x} \, dx$$

$$\begin{aligned} \int_0^\pi \frac{x \sin^3 x}{1 + \cos^2 x} \, dx &= \frac{\pi}{2} \int_0^\pi \frac{\sin^3 x}{1 + \cos^2 x} \, dx \\ &= \frac{\pi}{2} \int_0^\pi \frac{\sin^2 x \sin x}{1 + \cos^2 x} \, dx \\ &= \frac{\pi}{2} \int_0^\pi \frac{1 - \cos^2 x}{1 + \cos^2 x} \sin x \, dx \\ &= \frac{\pi}{2} \times 2 \int_1^{-1} \frac{1-u^2}{1+u^2} \, du \\ &= \pi \int_0^1 \frac{2 - (1+u^2)}{1+u^2} \, du \\ &= \pi \int_0^1 \frac{1-u^2}{1+u^2} \, du \\ &= \pi \left[2 \tan^{-1} u - u \right]_0^1 \\ &= \pi \left(2 \times \frac{\pi}{4} - 1 \right) \\ &= \frac{\pi}{2} (\pi - 2) \end{aligned}$$

$$\begin{aligned} \text{Let } u &= \cos x \\ du &= -\sin x \, dx \\ x=0, u &= 1 \\ x=\pi, u &= -1 \end{aligned}$$

1

1

1

(b) Let $x = 2 \tan \theta$

$$\frac{dx}{d\theta} = 2 \sec^2 \theta$$

$$\sqrt{x^2+4} = \sqrt{4 \tan^2 \theta + 4}$$

$$= 2 \sqrt{\tan^2 \theta + 1}$$

$$= 2 \sec \theta$$

↓

$$\therefore \int \frac{dx}{x^2 \sqrt{x^2+4}} = \int \frac{2 \sec^2 \theta}{4 \tan^2 \theta \cdot 2 \sec \theta} d\theta$$

$$= \int \frac{\sec \theta}{4 \tan^2 \theta} d\theta$$

$$= \frac{1}{4} \int \frac{1}{\cos \theta \times \frac{\sin^2 \theta}{\cos^2 \theta}} d\theta$$

$$= \frac{1}{4} \int \frac{\cos \theta}{\sin^2 \theta} d\theta$$

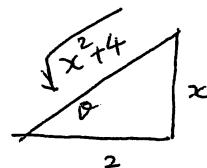
$$= -\frac{1}{4} \times \frac{1}{\sin \theta} + C$$

$$= -\frac{\sqrt{x^2+4}}{4x} + C$$

↓

↓

↓

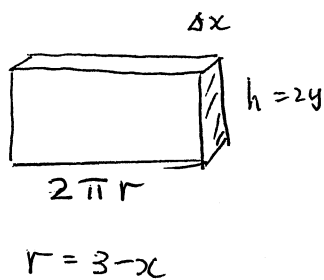
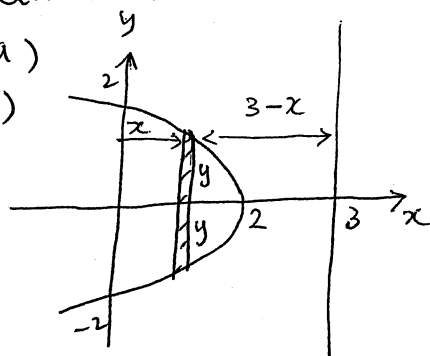


$$\sin \theta = \frac{x}{\sqrt{x^2+4}}$$

Question 4

(a)

i)



$$\Delta V = 2\pi r h \Delta x$$

$$= 2\pi(3-x) \times 2y \Delta x$$

$$= 4\pi(3-x)\sqrt{2-x} \Delta x$$

↓

$$\text{ii)} \quad V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^2 4\pi (3-x) \sqrt{2-x} \Delta x$$

$$= 4\pi \int_0^2 (3-x) \sqrt{2-x} dx \quad \downarrow$$

$$= -4\pi \int_2^0 (1+u) \sqrt{u} du \quad \downarrow \quad \text{Let } u=2-x$$

$$du = -dx$$

$$= 4\pi \int_0^2 \sqrt{u} + u\sqrt{u} du$$

$$x=0, u=2$$

$$x=2, u=0$$

$$= 4\pi \left[\frac{2}{3} u^{3/2} + \frac{2}{5} u^{5/2} \right]_0^2 \quad \downarrow$$

$$= 8\pi \left[u^{3/2} \left(\frac{1}{3} + \frac{u}{5} \right) \right]_0^2$$

$$= 8\pi \times 2\sqrt{2} \left(\frac{1}{3} + \frac{2}{5} \right)$$

$$= \frac{176\sqrt{2}}{15} \pi \text{ cubic units} \quad \downarrow$$

$$(b) \quad i) \quad \int_0^1 x^n \sqrt{1-x^2} dx = \int_0^1 x^{n-1} x \sqrt{1-x^2} dx$$

$$= \left[-\frac{1}{3} x^{n-1} (1-x^2)^{3/2} \right]_0^1 + \frac{n-1}{3} \int_0^1 x^{n-2} (1-x^2) \sqrt{1-x^2} dx \quad \downarrow$$

$$= 0 + \left(\frac{n-1}{3} \right) \int_0^1 \{ x^{n-2} \sqrt{1-x^2} - x^n \sqrt{1-x^2} \} dx \quad \downarrow$$

$$= -\left(\frac{n-1}{3} \right) I_n + \left(\frac{n-1}{3} \right) I_{n-2}$$

$$\left(1 + \frac{n-1}{3} \right) I_n = \left(\frac{n-1}{3} \right) I_{n-2}$$

$$I_n = \frac{n-1}{n+2} I_{n-2} \quad \downarrow$$

$$ii) \quad I_0 = \int_0^1 \sqrt{1-x^2} dx$$

$$= \frac{\pi}{4} \quad \downarrow \quad \left(\frac{1}{4} \text{ of a circle, radius } 1 \right)$$

$$I_2 = \frac{1}{4} I_0$$

$$= \frac{\pi}{16}$$

$$I_4 = \frac{3}{8} I_2$$

$$= \frac{\pi}{32} \quad \downarrow$$